Document Recommender System — Technical Report

TF-IDF content-based recommender & scikit-learn evaluation · Author: Felix (@C0z1) · November 2026

1. Description of the approach

The system is a **content-based recommender** built on the **TF-IDF + cosine similarity** approach (Path A), chosen for its simplicity and transparency: every recommendation score can be traced directly to the words the documents share. The corpus holds **33 documents** across five categories — Policy/Process, Support/Incidents, FAQ, Meeting Minutes and News/Article — stored in `corpus.csv` with the columns `id, title, text, category`.

Each document's *title + text* is lowercased, stripped of punctuation and vectorized with `TfidfVectorizer` (English stop-words, unigrams and bigrams). Similarity between any two documents — or between a free-text query and every document — is the cosine of their vectors. The recommender ranks all documents and returns the **Top-5**, excluding the seed itself. Two input modalities are supported: a **seed document ID** (`--id DOC011`) and a **free-text query** (`--query "..."`).

The results are coherent. A VPN support ticket retrieves the VPN FAQ as its top match (score 0.29), and the free-text query *"I forgot my password and I am locked out"* returns the account-recovery FAQ (0.41) and the password-reset ticket (0.24) — exactly the documents a help desk would reach for.

2. Evaluation summary

Quality was measured with scikit-learn, defining relevance automatically: a document is relevant when it shares the seed's category. For the seed `DOC011` (VPN Disconnection) at Top-5:

Metric	Result	Meaning
Precision	0.60	3 of the 5 returned documents were relevant (low noise).
Recall	0.43	3 of the 7 relevant documents were found (partial coverage).
F1-score	0.50	Balanced score; good precision limited by coverage.

Confusion matrix: TP 3 FP 2 FN 4 TN 23. The system favours **precision over recall** — expected, since Top-5 cannot cover all 7 relevant documents. It returns a short, mostly-correct list rather than a wide one, which builds user trust at the cost of some missed knowledge.

3. Development areas of opportunity (improvements)

- **Semantic embeddings (Path B).** TF-IDF matches words, not meaning, so synonyms ("login" vs "sign-in") and paraphrases are missed. A `sentence-transformers` model would raise recall.
- **Metadata & role-based filters.** Adding product, recency, author and permissions would raise precision and ensure users only see documents they are allowed to access (security by design).
- **Adaptive result size.** A score threshold or a tuned *K* would let the retrieved-set size adapt to the query instead of always returning five.
- **Hybrid retrieval & query expansion.** Combining lexical (TF-IDF) and semantic signals, plus expanding queries with related terms, handles ambiguity and short documents better.
- **Human-labeled ground truth & continuous evaluation.** Category-based relevance is a convenient but pessimistic yardstick (the VPN FAQ, genuinely useful, is counted as a false positive). Human labels and ongoing metric tracking would give a fairer, drift-aware measure of quality as the corpus grows.

4. Organizational reflection — video

> **Reflection video link:** *[PASTE YOUR VIDEO LINK HERE before submitting]*
> The script/talking points (why this reduces rework, the limitations, and what to improve for production) and the infographic content are in `ANALYSIS.md`, sections A.2 and B.4.

Full code, dataset and test evidence: see the repository README and the `evidencias/` folder. Recommender: `recommender.py` · Evaluation: `evaluate.py`.